ADVANCED AI FOR BUSINESS AUTOMATION · FORMATIVE ASSESSMENT

# Agentic AI for the Client Advisory Process

Onboarding · Risk profiling · Recommendation · Compliance check · Review

[Team members] · [Date of viva]

Academic prototype · synthetic data · not financial advice

THE PROBLEM

# Manual advisory takes about 90 minutes a case, and every rule failure restarts work

### Slow

Analysts move each case by hand through 8 steps and redo calculations in spreadsheets.

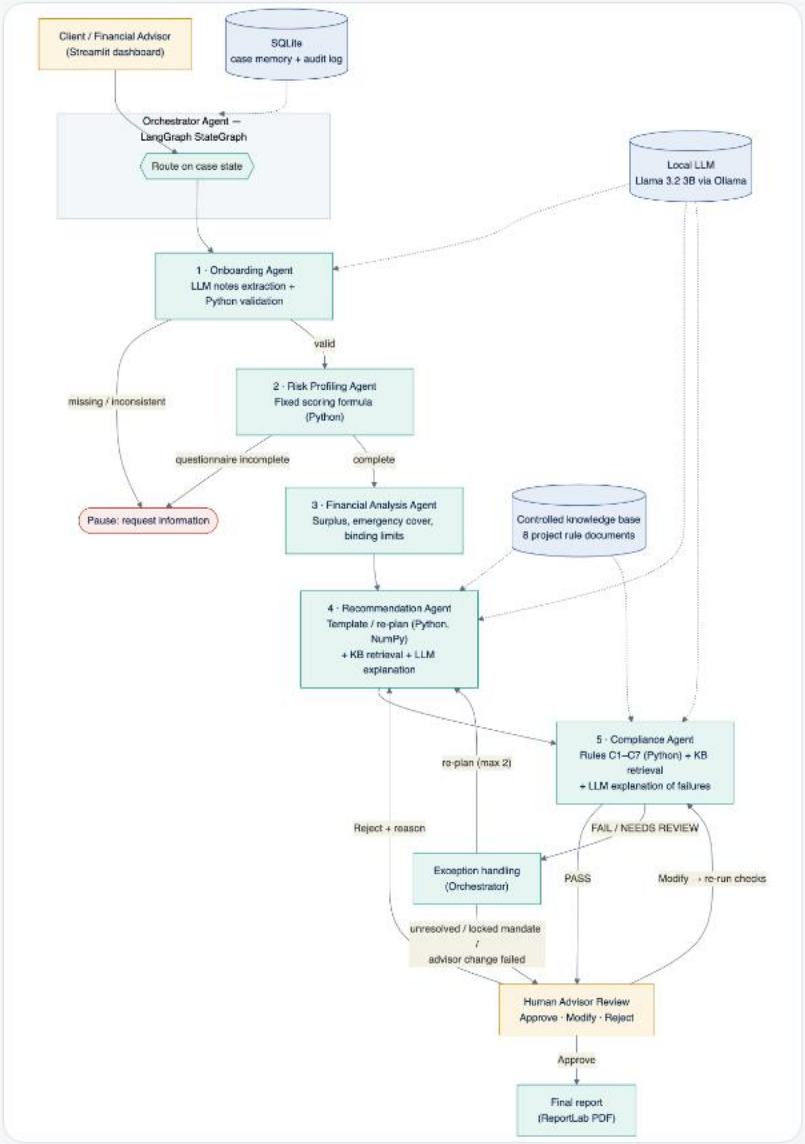
### Late exceptions

Missing data and rule breaches surface at compliance, costing about a day per rework hand-off.

### Weak audit trail

Decision evidence is spread across notes, email and spreadsheets.

Baseline figures are illustrative assumptions, as allowed by the project guidelines · 2




THE SOLUTION

# One orchestrator routes each case between five specialist agents

- Agents write structured results to one shared case state
- The orchestrator decides: continue, pause, re-plan or escalate
- A human advisor approves, modifies or rejects every case
- Every step lands in a SQLite audit log

THE AGENTS

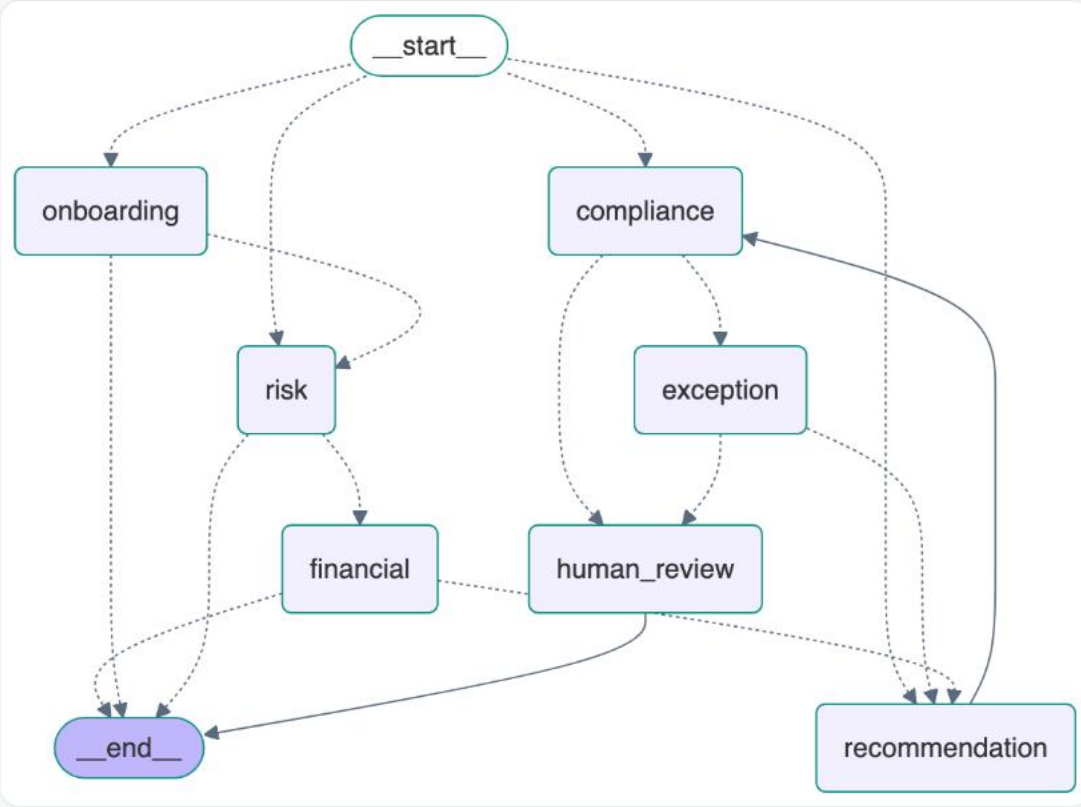
# Each agent owns one job and one output

| Agent | Responsibility | Tools |
|---|---|---|
| Orchestrator | Routes, re-plans, escalates, briefs the advisor | LangGraph, SQLite, LLM |
| Onboarding | Reads notes; checks completeness and consistency | LLM, Python |
| Risk Profiling | Scores 6 answers with a fixed formula | Python |
| Financial Analysis | Surplus, emergency cover, binding limits | Python |
| Recommendation | Proposes and revises the allocation; explains it | Python, NumPy, retrieval, LLM |
| Compliance | Checks rules C1–C7 independently | Python, retrieval, LLM |

ORCHESTRATION

# On a failed check, the orchestrator re-plans or escalates

1. Only “needs review” items → escalate
2. Advisor’s own change failed → escalate
3. Client mandate locked → escalate with a compliant alternative
4. Two re-plans already tried → escalate
5. Otherwise → send the violated rules back and re-plan




Graph exported directly from the code: dotted edges are conditional routes · 5

AI AND RULES

# The LLM writes the words; Python makes every decision

## Local LLM (Llama 3.2 3B)

- Extracts client fields from free-text notes
- Explains why an allocation fits
- Explains failed rules in plain language
- Writes the advisor briefing

## Deterministic Python

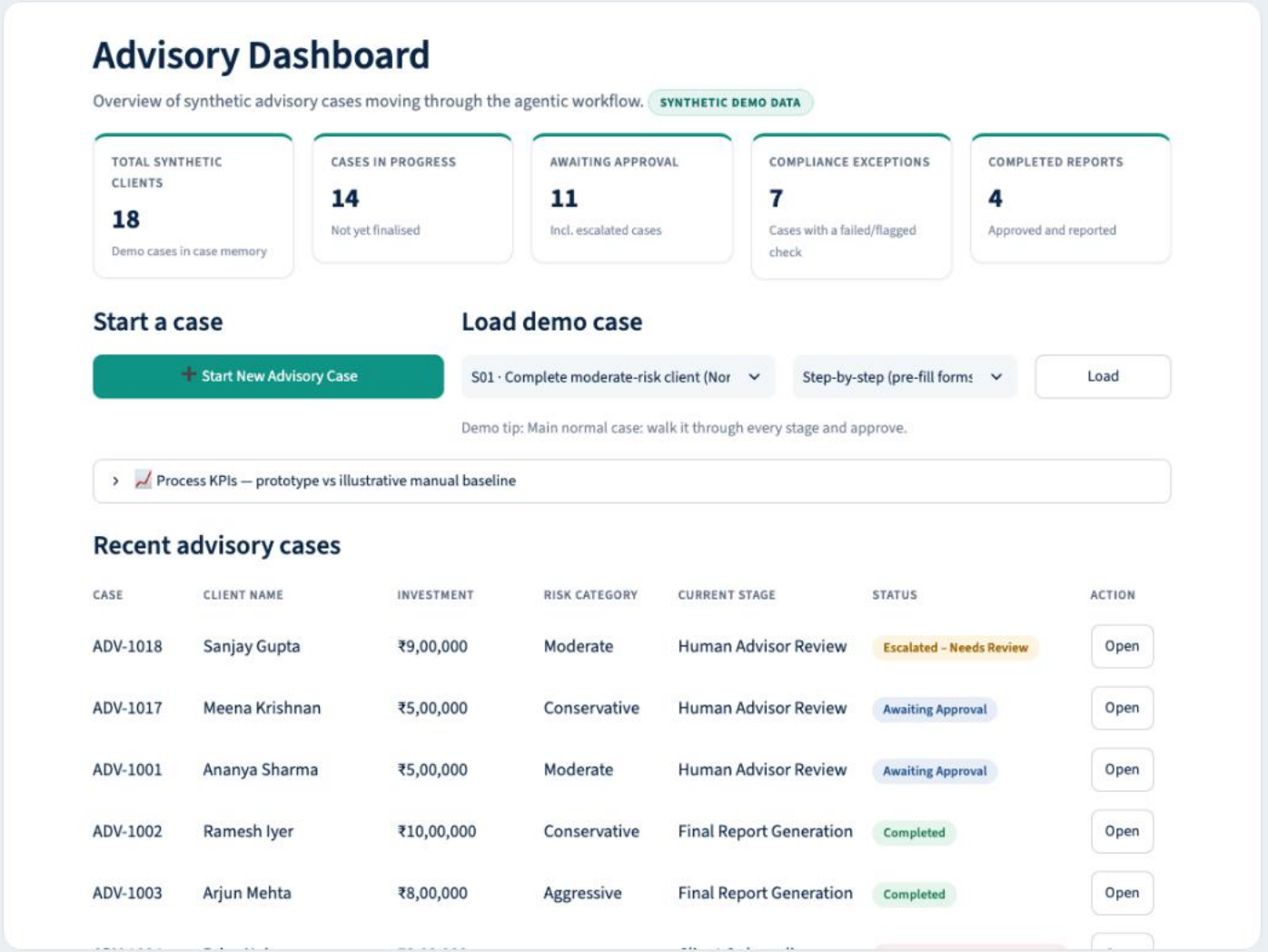
- Risk score (6–30) and category
- Allocation, rupee amounts, projections
- PASS / FAIL for rules C1–C7
- Validation of every LLM-extracted value

Runs free on a laptop via Ollama; no client text leaves the machine; template fallback if the LLM is down · 6

LIVE DEMO · THE DASHBOARD

# 16 synthetic cases, each at a visible stage

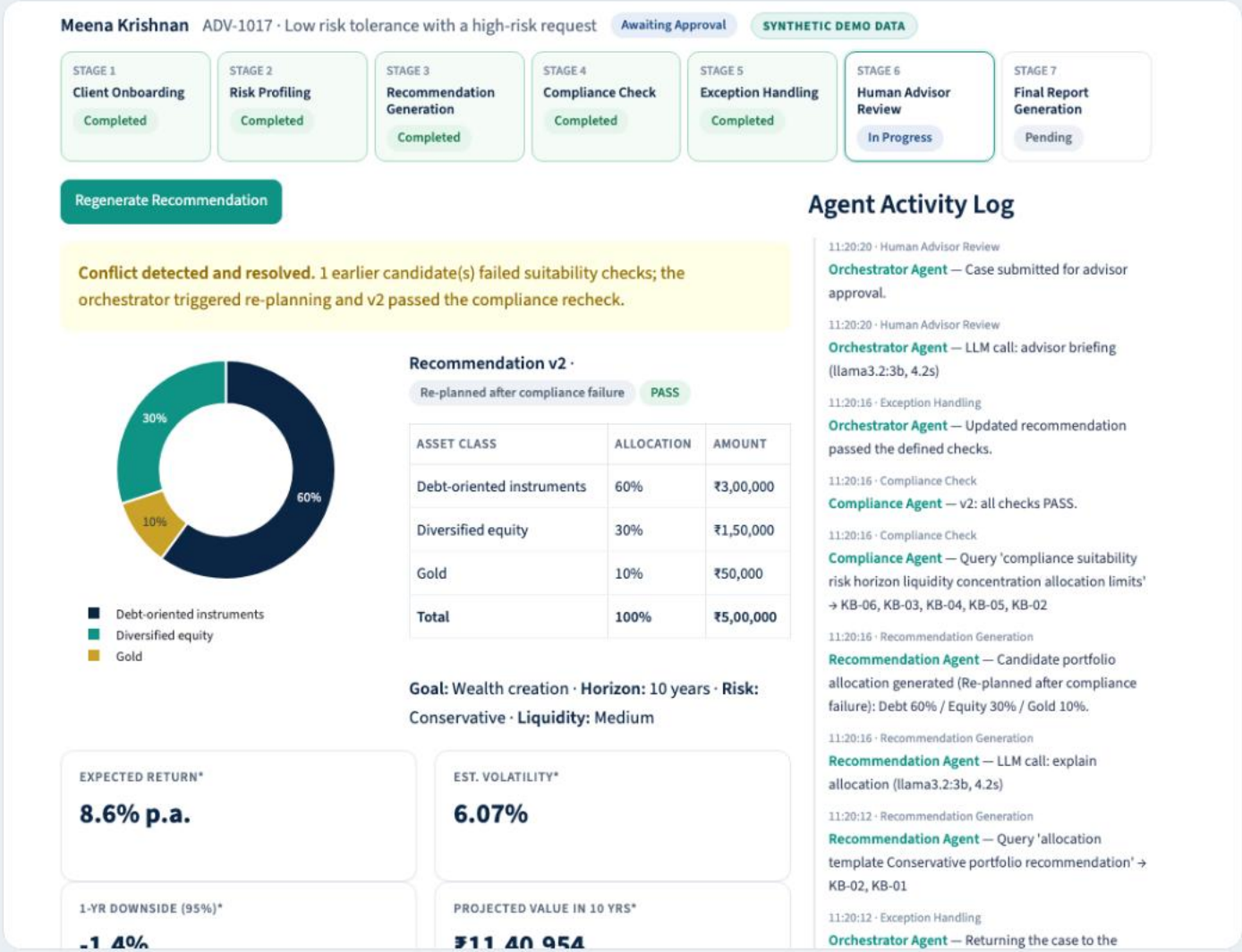
- Start a case or load a demo scenario
- Paste client notes; the LLM fills the form
- Walk a normal case to an approved PDF



EXCEPTION · RE-PLAN (S06)

# A risk mismatch fixed in 14 seconds, no human needed

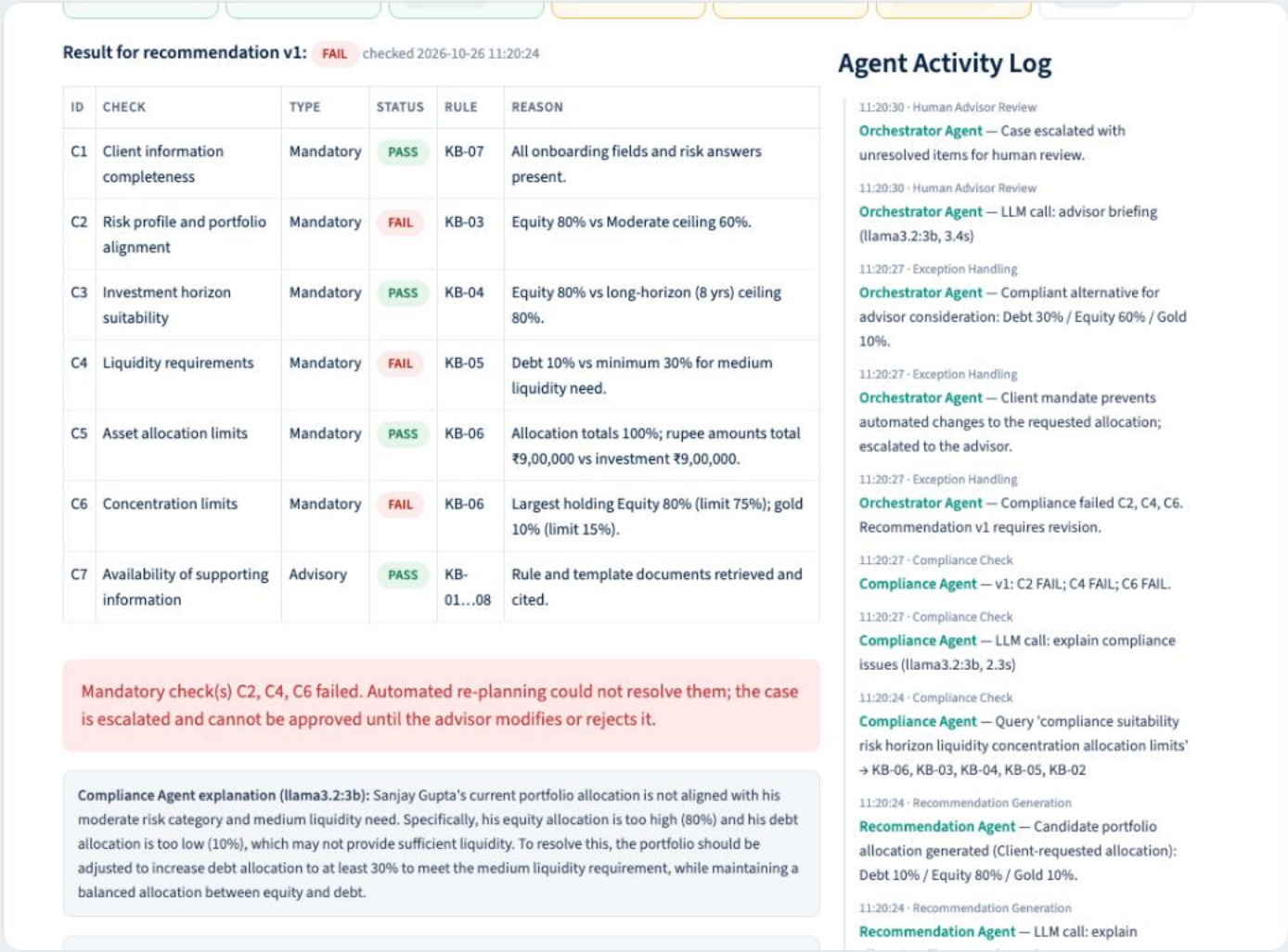
1. Conservative client asks for 70% equity
2. Compliance fails C2 and C4
3. Orchestrator re-plans to 30% equity, 60% debt
4. v2 passes; advisor gets an LLM briefing



EXCEPTION · ESCALATION (S09)

# When rules can't be met automatically, a person decides

- Client mandate locks 80% equity
- C2, C4 and C6 fail; case escalated
- LLM explains the failures; a compliant alternative is prepared
- Approve stays disabled until the advisor modifies or rejects



**Result for recommendation v1:** FAIL checked 2026-10-26 11:20:24

| ID | CHECK | TYPE | STATUS | RULE | REASON |
|---|---|---|---|---|---|
| C1 | Client information completeness | Mandatory | PASS | KB-07 | All onboarding fields and risk answers present. |
| C2 | Risk profile and portfolio alignment | Mandatory | FAIL | KB-03 | Equity 80% vs Moderate ceiling 60%. |
| C3 | Investment horizon suitability | Mandatory | PASS | KB-04 | Equity 80% vs long-horizon (8 yrs) ceiling 80%. |
| C4 | Liquidity requirements | Mandatory | FAIL | KB-05 | Debt 10% vs minimum 30% for medium liquidity need. |
| C5 | Asset allocation limits | Mandatory | PASS | KB-06 | Allocation totals 100%; rupee amounts total ₹9,00,000 vs investment ₹9,00,000. |
| C6 | Concentration limits | Mandatory | FAIL | KB-06 | Largest holding Equity 80% (limit 75%); gold 10% (limit 15%). |
| C7 | Availability of supporting information | Advisory | PASS | KB-01…08 | Rule and template documents retrieved and cited. |

Mandatory check(s) C2, C4, C6 failed. Automated re-planning could not resolve them; the case is escalated and cannot be approved until the advisor modifies or rejects it.

**Compliance Agent explanation (llama3.2:3b):** Sanjay Gupta's current portfolio allocation is not aligned with his moderate risk category and medium liquidity need. Specifically, his equity allocation is too high (80%) and his debt allocation is too low (10%), which may not provide sufficient liquidity. To resolve this, the portfolio should be adjusted to increase debt allocation to at least 30% to meet the medium liquidity requirement, while maintaining a balanced allocation between equity and debt.

## Agent Activity Log

11:20:30 · Human Advisor Review
**Orchestrator Agent** — Case escalated with unresolved items for human review.

11:20:30 · Human Advisor Review
**Orchestrator Agent** — LLM call: advisor briefing (llama3.2:3b, 3.4s)

11:20:27 · Exception Handling
**Orchestrator Agent** — Compliant alternative for advisor consideration: Debt 30% / Equity 60% / Gold 10%.

11:20:27 · Exception Handling
**Orchestrator Agent** — Client mandate prevents automated changes to the requested allocation; escalated to the advisor.

11:20:27 · Exception Handling
**Orchestrator Agent** — Compliance failed C2, C4, C6. Recommendation v1 requires revision.

11:20:27 · Compliance Check
**Compliance Agent** — v1: C2 FAIL; C4 FAIL; C6 FAIL.

11:20:27 · Compliance Check
**Compliance Agent** — LLM call: explain compliance issues (llama3.2:3b, 2.3s)

11:20:24 · Compliance Check
**Compliance Agent** — Query 'compliance suitability risk horizon liquidity concentration allocation limits' → KB-06, KB-03, KB-04, KB-05, KB-02

11:20:24 · Recommendation Generation
**Recommendation Agent** — Candidate portfolio allocation generated (Client-requested allocation): Debt 10% / Equity 80% / Gold 10%.

11:20:24 · Recommendation Generation
**Recommendation Agent** — LLM call: explain

TESTING

# Every scenario behaved as designed

**16/16**
scenarios reached the expected outcome

**8**
exception scenarios, against a minimum of 3

**45**
automated tests passing

Covered: missing data, incomplete questionnaire, risk mismatch, concentration and liquidity breaches, locked mandate, knowledge-source outage, advisor approve / modify / reject, and what-if changes.

PROCESS KPIS

# Processing time falls from 90 to about 10 minutes a case

| KPI | Manual (assumed) | Prototype (measured) |
|---|---|---|
| Time per case | 90 min | About 10 min, mostly advisor review (–89%) |
| Missing data | Found late | Caught at intake, 3 of 3 |
| Rework cycle | About 24 h per hand-off | Seconds; 4 of 6 auto-resolved |
| Rule consistency | Varies by analyst | 100% identical results |
| Human approval | Manual sign-off | 100%; failing cases blocked |
| Audit trail | Notes and spreadsheets | Every action and decision logged |

Baselines and the 10-minute review time are illustrative assumptions; prototype values are measured · 11

GOVERNANCE

# No AI component can approve a case or override a rule

### Human in the loop

Approve is disabled while a mandatory check fails; “needs review” items require a written justification.

### Re-checks on change

Advisor modifications re-run compliance; rejections need a reason and return the case for revision.

### Privacy

Synthetic data only; the LLM runs locally, so no text leaves the laptop.

### Full audit trail

Every agent action, tool call, LLM call, exception and decision is timestamped in SQLite.

LIMITS AND NEXT STEPS

# A working prototype, with a clear path to a pilot

## Limitations today

- Synthetic data and simplified rules
- Baseline KPIs are assumptions
- Small 3B model can misread vague notes
- Keyword retrieval over 8 documents
- No login or user roles

## Next steps

- Vector search over real product and regulatory documents
- Enterprise-approved LLM with an accuracy test set
- Advisor and compliance roles with authentication
- Pilot with one team to measure a real baseline

Lesson learned: do deterministic work before the LLM, e.g. converting “16 lakh” to rupees in Python · 13

SUMMARY

# Agents run the process. People make the decision.

Five agents and an orchestrator handled all 16 scenarios, re-planned when rules allowed, escalated when they did not, and logged every step.

**Questions?**

github.com/raghul-t2/AI-Automation